

# Arnesh Banerjee

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## EDUCATION

**Heritage Institute of Technology, Kolkata**

*Bachelor of Technology in Data Science*

GPA: 8.876/10

*Aug. 2023 – June. 2027*

## RESEARCH INTERESTS

Multi-Agent Reinforcement Learning (MARL) · RL Environments & Simulation · AI Safety & Alignment · Deep Learning · Applied ML · LLMs · AI

## RESEARCH EXPERIENCE

**IIT Kharagpur** (Summer Internship 2026, Upcoming)

Starting 23rd May 2026

*Advisor: Dr. Sourangshu Bhattacharya*

*On Site*

- Selected for the **GRISHMA Summer Internship Program** at IIT Kharagpur via the official KRITI portal (kriti.iitkgp.ac.in); will work with Dr. Sourangshu Bhattacharya on building India's first genomic foundational model (IgLM), a population-specific genomic language model pretrained on GenomeIndia, IndiGen, and GenomeAsia using the StripedHyena2 architecture.

**Jadavpur University**

November 2025 – Present

*Advisor: Prof. Debotosh Bhattacharjee*

*Remote*

- Achieved 98.10% Dice and 96.39% IoU on breast thermography tumor segmentation using a ResNet34 + ASPP + ViT decoder with SE attention, evaluated via 5-fold cross-validation.
- Conducted a rigorous ablation study (20 training runs) across U-Net, ResNet-only, and hybrid architectures, demonstrating that simpler models can match complex designs and highlighting generalization trade-offs in medical imaging.
- Built a fully automated cross-validation pipeline with checkpointing, experiment logging, and reproducible training, producing publication-ready comparative metrics for medical image analysis.

**New Jersey Institute of Technology**

June 2025 – July 2025, August 2025 - November 2025

*Advisor: Prof. Arnob Ghosh*

*On Site, Virtual*

- Constructed a cost-model training dataset for a modified Safe RLHF pipeline leading to improved results in the cost-model.
- Designed and tested comparative 800 datapoint safety benchmark against state-of-the-art LLMs (GPT, DeepSeek, Gemini, Qwen, Mistral).
- Demonstrated 50% higher safety compliance than ChatGPT 5 across benchmark scenarios.

**Heritage Institute of Technology**

October 2024 – March 2025

*Advisor: Prof. Arpita Talukdar*

*On Site*

- Developed a dual-stage ML framework for breast cancer diagnosis and recurrence prediction using WDBC and WPBC datasets, evaluating RF, SVM, Logistic Regression, MLP, and XGBoost with stratified 10-fold cross-validation.
- Implemented RFE and SFS feature selection with SMOTE and GridSearchCV tuning, improving WPBC accuracy to 93.67% (SVM + RFE) and WDBC to 97.77% (LogReg + RFE).
- Conducted comparative performance analysis and interpreted clinically relevant nuclear features, reporting stable cross-validated results with confidence intervals.

## PUBLICATIONS

- Ayushi Bhattacharjee, **Arnesh Banerjee**, Arpita Talukdar. 2026. Recursive and Wrapper-Based Feature Selection for Breast Cancer Diagnosis and Prognosis. In the proceedings of the 4th Analytics Global Conference (AGC 2026), March 2026. **Oral**.

## PRE-PRINTS

- Kartik Pandit, Sourav Ganguly, **Arnesh Banerjee**, Shaahin Angizi, Arnob Ghosh. 2025. Certifiable Safe RLHF: Fixed-Penalty Constraint Optimization for Safer Language Models. arXiv preprint arXiv:2510.03520.
- Arnesh Banerjee**, Debotosh Bhattacharjee. Multiscale Transformer-Enhanced U-Net with Level-Set Regularization for Breast Thermography Segmentation. *Advisor: Prof. Debotosh Bhattacharjee*

## INDEPENDANT & ONGOING RESEARCH

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- **Arnesh Banerjee**. Co-evolutionary Multi Agent RL for Autonomous Drones. In collaboration with the AI for Defence Lab, ULiège, Belgium.
- Kartik Pandit, Sourav Ganguly, **Arnesh Banerjee**, Ayushi Bhattacharjee, Avirup Chakraborty, Arnob Ghosh. Analyzing Historical Revisionism in LLMs in the Context of Indian History. *Advisor: Dr. Arnob Ghosh.*
- **Arnesh Banerjee**, Ayushi Bhattacharjee, Subhajit Datta. Understanding the Limitations of LLMs in Mathematical Reasoning. *Advisor: Prof. Subhajit Datta.* Part of B.Tech degree coursework.

## ACHIEVEMENTS

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- **Department Third:** 4th Semester. SGPA: 9.46. B.Tech CSE(DS), Heritage Institute of Technology, Kolkata
- **Institutional Innovation Council(IIC):** Selected one of 10 IIC members across all years to represent the Department of CSE(DS), HIT, Kolkata
- **Paper Presenter:** Presenting Author for paper at AGC 2026 on Applied ML in Medical domain.
- **WBJEE Qualified:** Top 5.3% in the state on WBJEE, 2023

## TECHNICAL SKILLS

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**Languages:** Python, R, C, Java, SQL, LaTeX

**Machine Learning:** PyTorch, Scikit-Learn, Keras, H2O, CatBoost, XGBoost, Reinforcement Learning, Deep Q-Networks (DQN), Double DQN, PPO, SAC

**Simulation & RL Environments:** Gymnasium, OpenAI Gym, Stable-Baselines3, JAX

**Data & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly

**Frameworks & Tools:** Flask, FastAPI, Shiny, Docker, Git

**Relevant Courses:** Operating Systems, Database Management System, Data Structures, Algorithms, Machine Learning, Data Mining, Data Warehousing